FACE EMOTION CLASSIFICATION

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PROBLEM STATMENT

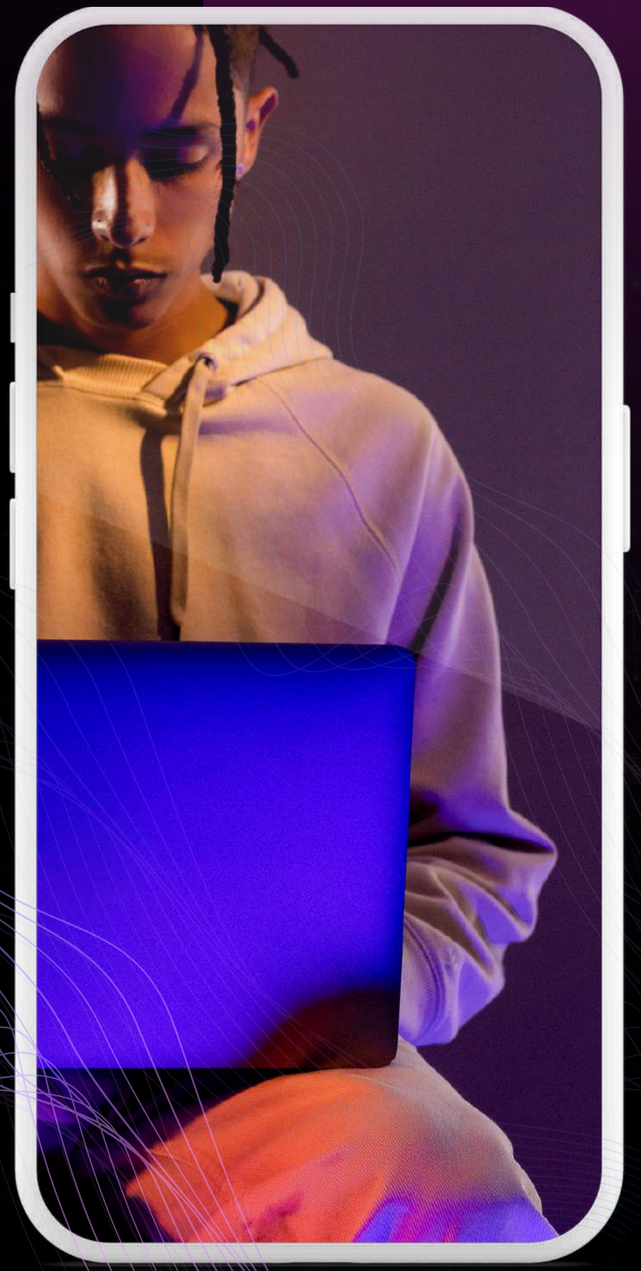
- The AffectNet dataset is one of the largest and most comprehensive datasets available for facial expression recognition, used primarily for emotion recognition tasks. It contains manually labeled facial images that represent a wide range of emotions

- All images are exactly 96x96 pixels.(On Kaggle) (approx. 28,000 images)

- The dataset consists of facial images in a wide variety of poses, lighting conditions, and occlusions (e.g., glasses, facial hair, head coverings).

- There are 8 classes (7 basic + 1 other emotions) in this dataset for classification.

  (a) Disgust (b) Fear
  (c) Neutral (d) Happy
  (e) Contempt (f) Anger
  (g) Sad (h) Surprise



VISION & MISSION

PRPROCESSING

> Removing the ambigous Data
> Analysing the Data.
> Structuring the Data.
> Train Test Split.

AUGMENTATION

> Random Rotation
> Random Hue & Contrast
> Random Crop + Resize
> Random Flip

BASE MODEL

> Using Pretrained Conv Base Models
- EfficientNet-B0
- DensceNet-121
- VGGFace

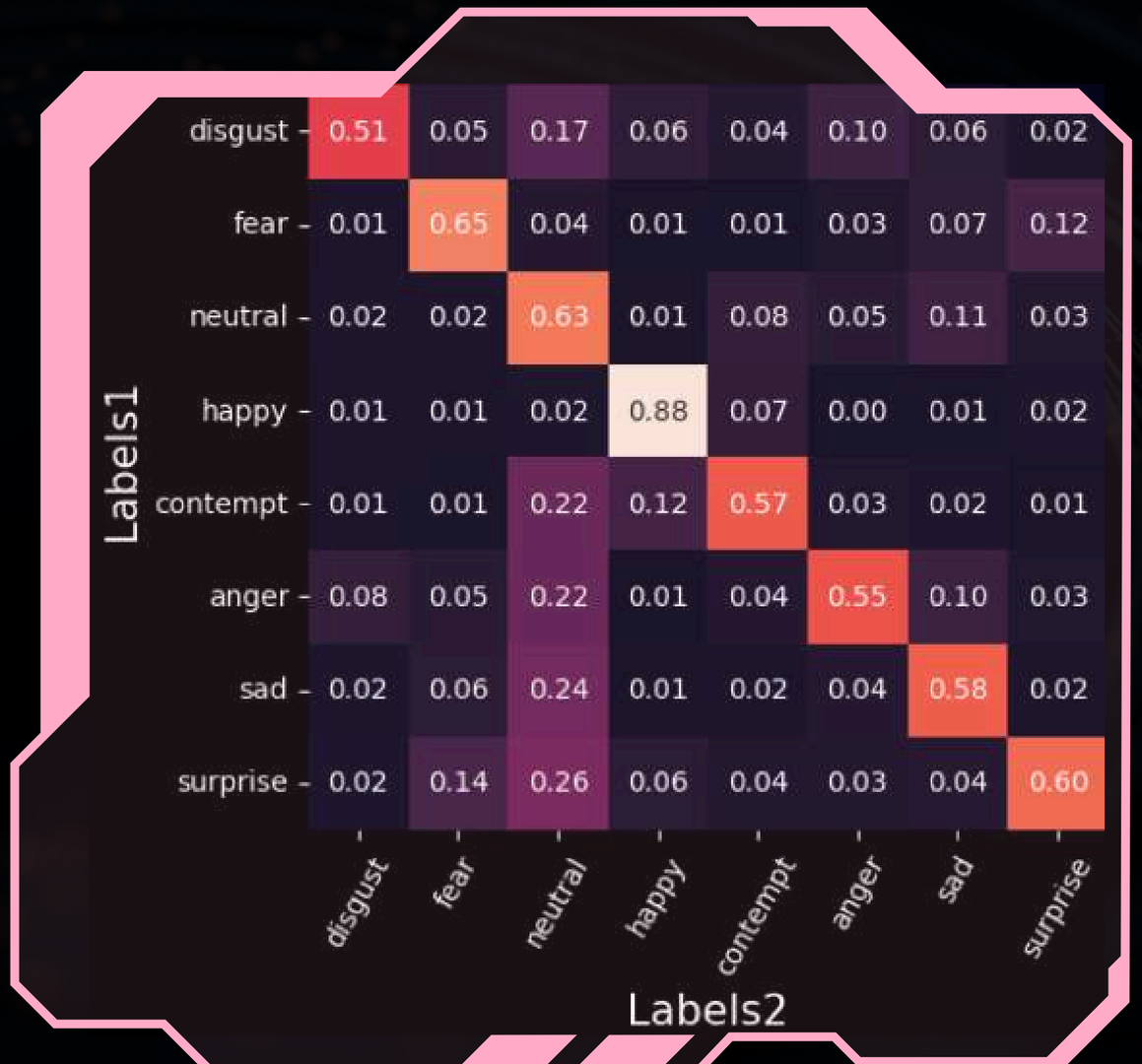
CLASSIFIER MODEL

> Using Ensemble Classifier (varying activation)
- Relu
- Gelu
- Swish

COMPARNING MODELS

Comparing the result from the three individual model and their ensembled model.

CORRELATION AMONG TWO LABELS



Removing Ambigous Data

	Class	label_count
0	disgust	1780
1	fear	1967
2	neutral	1798
3	happy	3820
4	contempt	1842
5	anger	1998
6	sad	1729
7	surprise	2771
8	Total	17705

SAMPLE IMAGES...

DISGUST

FEAR

NEUTRAL

HAPPY

CONTEMPT

ANGER

SAD

SURPRISE

AUGMENTATION ON SAMPLE

BASE MODELS

01 EFFIECIENTNET

EfficientNet excels at balancing accuracy and computational efficiency, making it ideal for large-scale datasets like AffectNet with minimal resource usage.

02 DENSENET

DenseNet offers strong feature reuse through dense connections, making it effective for learning subtle facial expressions in the AffectNet dataset.

03 VGGFace

VGGFace, pretrained on facial data, is highly specialized for face recognition tasks, making it well-suited for facial expression recognition on AffectNet

ENSEMBLE

CLASSIFIER ENSEMBLE :

Output from the base model was passed to the the three classifier (FNN) model with same artitecture but varying activations and the output from them was concatnated and then passed t o ouput layer.

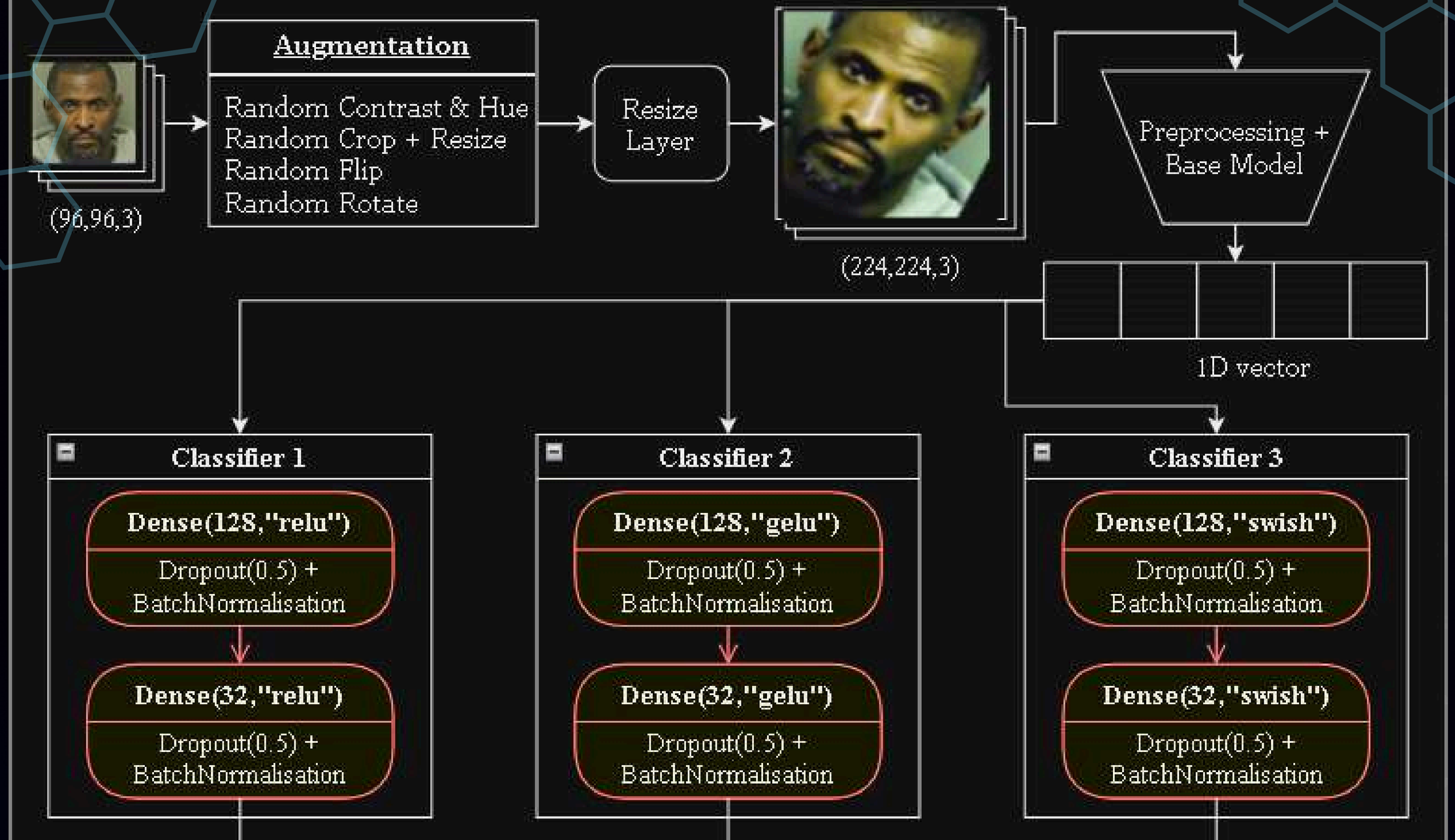
MODEL ENSEMBLE :

The ensemble of the three individual models was created by concatenating their outputs and passing the combined result through the final output layer.

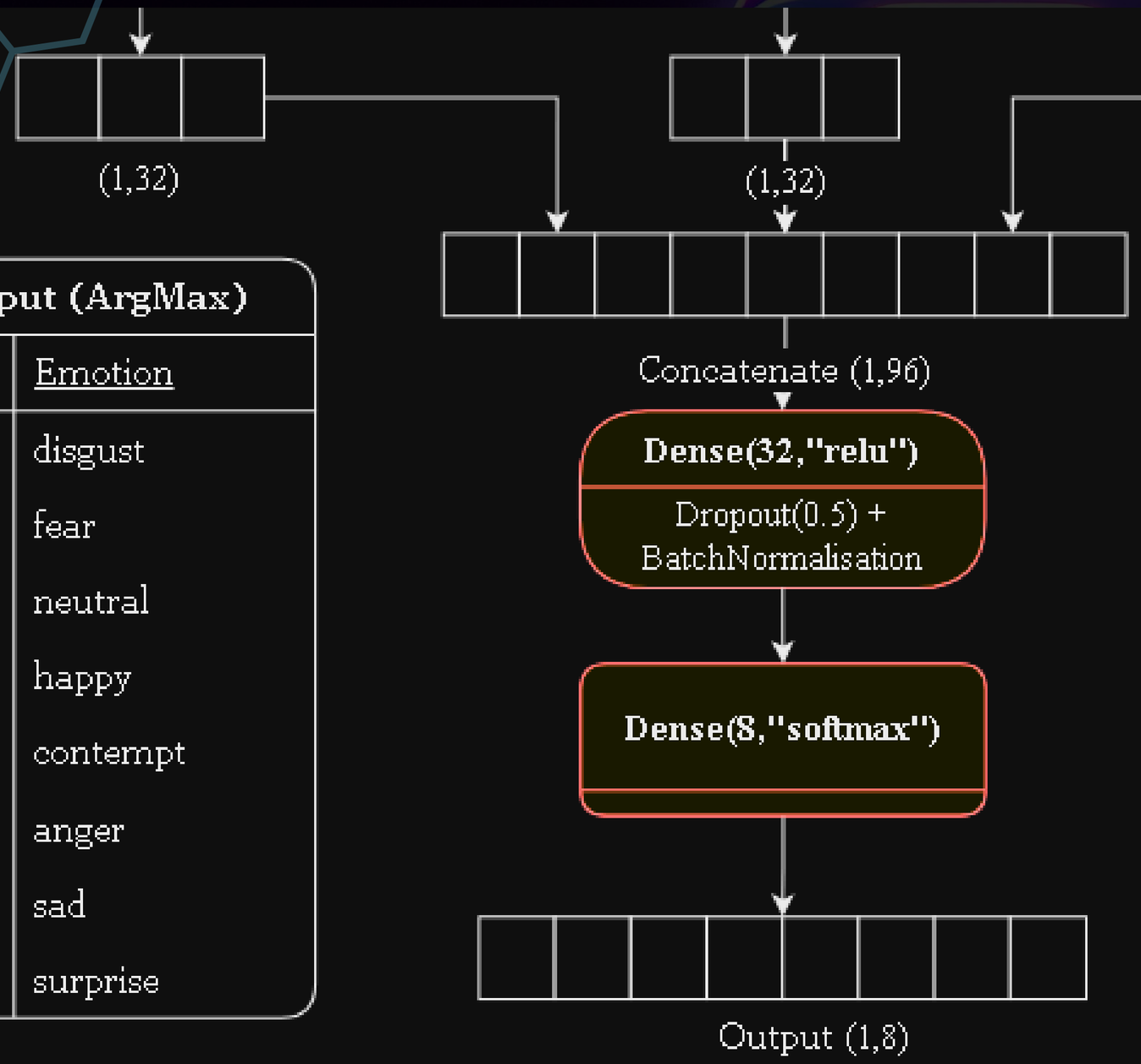
MODEL PARAMETER

- Image_Size : (96,96,3)
- Batch_Size : 32
- Number of Epochs and Leanring rate (in accordance to the base model).

INDIVIDUAL MODEL



CONTINUED...



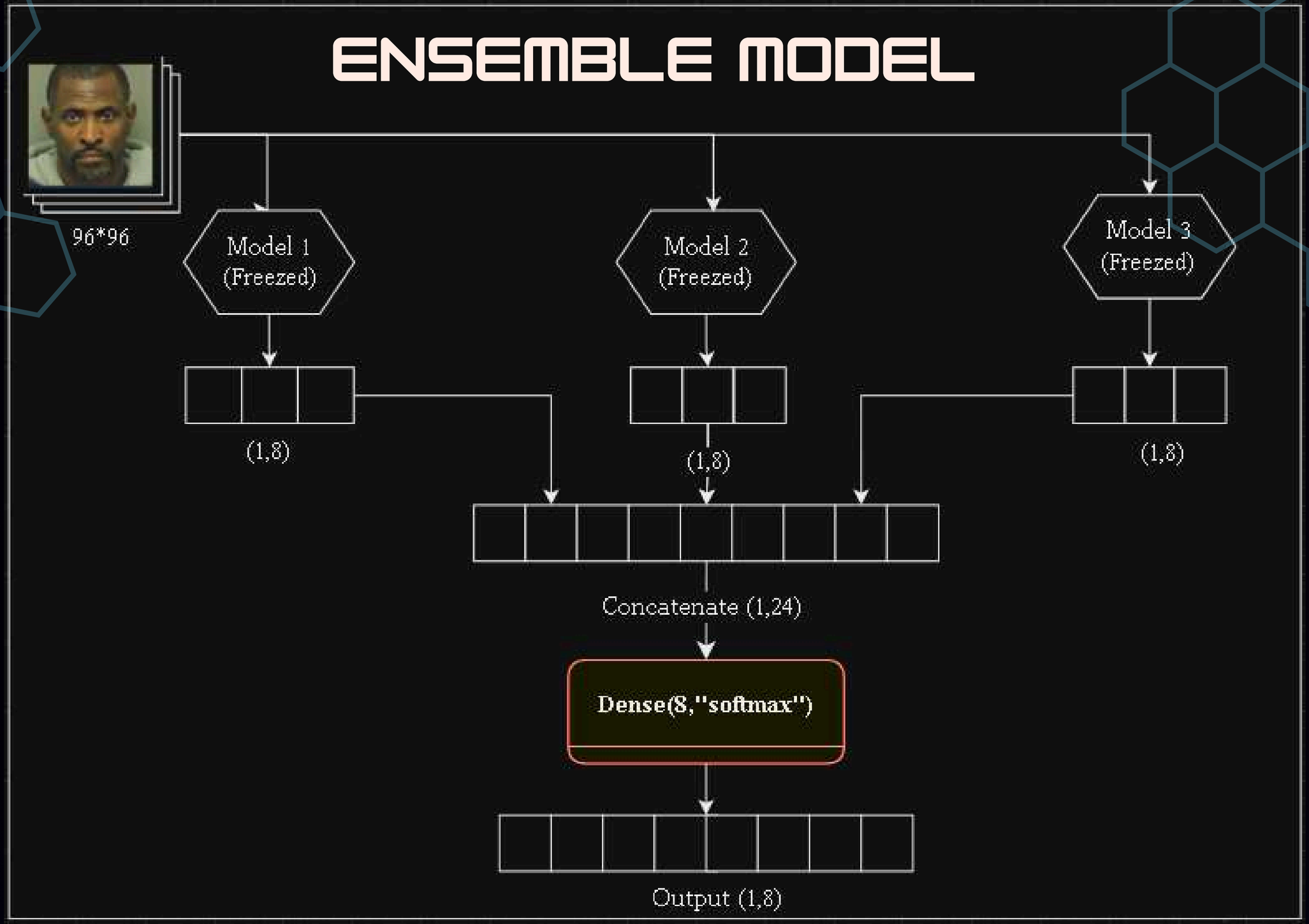
Output (ArgMax)

Label	Emotion
0	disgust
1	fear
2	neutral
3	happy
4	contempt
5	anger
6	sad
7	surprise

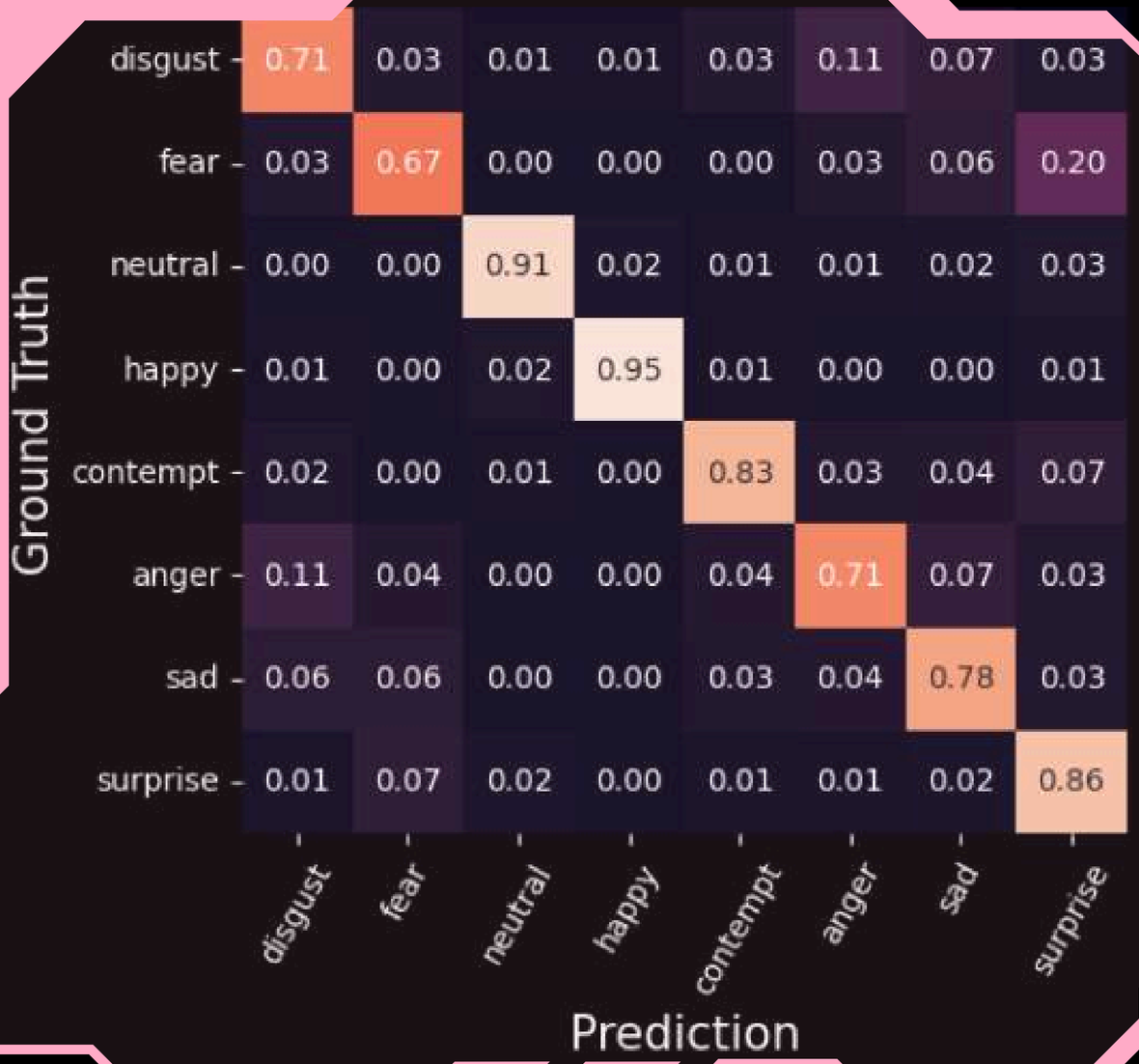
Base Models

> EfficientNet-B0
> DenseNet-121
> VGGFace

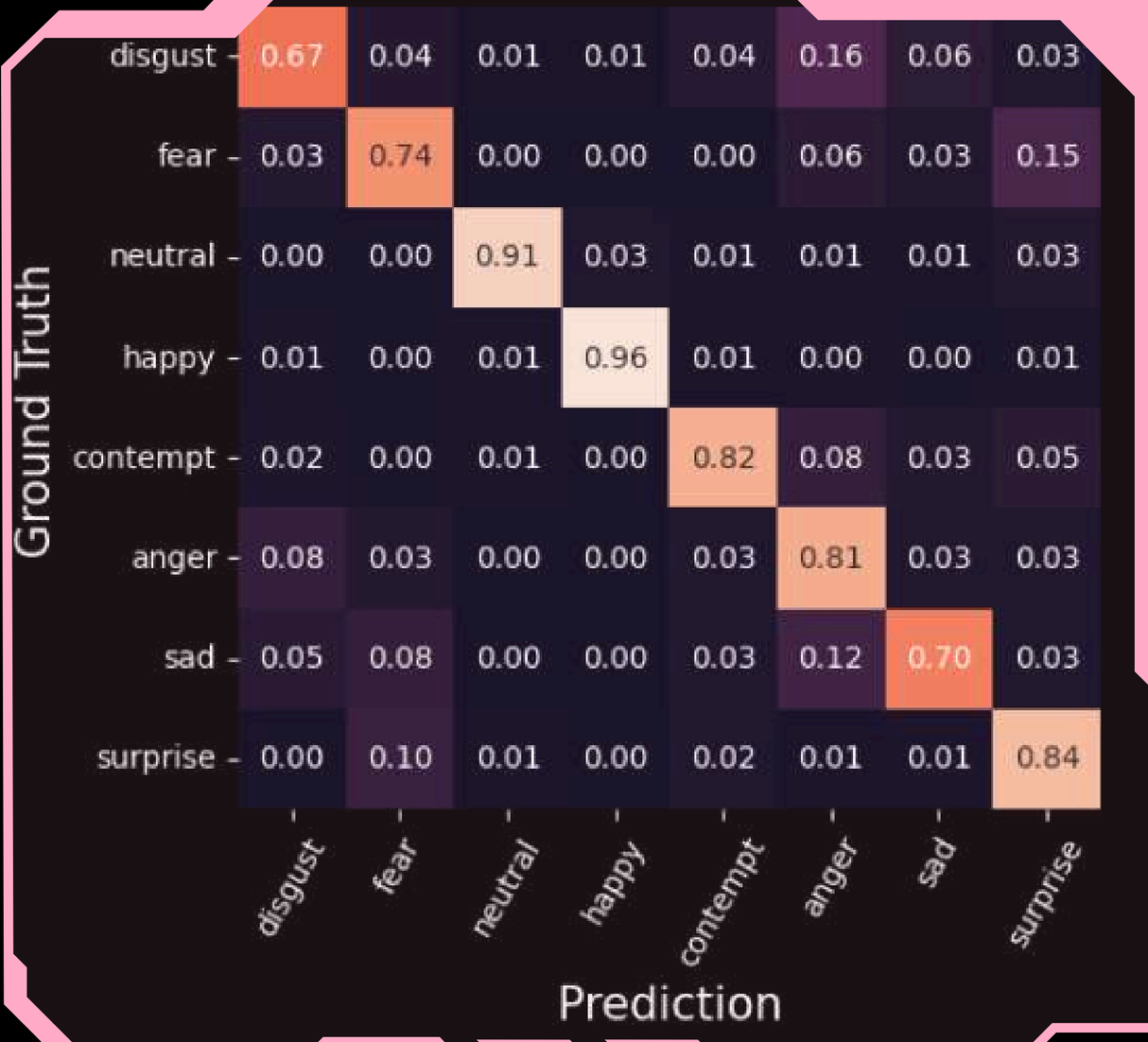
ENSEMBLE MODEL



EFFICIENET-B0



DENSENET-121



VGGFACE

Ground Truth \ Prediction	disgust	fear	neutral	happy	contempt	anger	sad	surprise
disgust	0.744	0.037	0.003	0.006	0.065	0.104	0.025	0.017
fear	0.036	0.718	0.003	0.000	0.005	0.046	0.043	0.150
neutral	0.006	0.006	0.772	0.025	0.089	0.036	0.028	0.039
happy	0.010	0.000	0.010	0.929	0.038	0.000	0.003	0.009
contempt	0.024	0.003	0.008	0.005	0.870	0.027	0.011	0.052
anger	0.117	0.022	0.007	0.000	0.065	0.748	0.035	0.005
sad	0.078	0.040	0.003	0.000	0.058	0.055	0.725	0.040
surprise	0.016	0.094	0.014	0.002	0.045	0.025	0.018	0.785

ENSEMBLE

Ground Truth \ Prediction	disgust	fear	neutral	happy	contempt	anger	sad	surprise
disgust	0.75	0.03	0.00	0.01	0.04	0.11	0.04	0.02
fear	0.02	0.76	0.00	0.00	0.00	0.03	0.04	0.15
neutral	0.00	0.00	0.92	0.02	0.01	0.01	0.02	0.02
happy	0.00	0.00	0.01	0.98	0.00	0.00	0.00	0.01
contempt	0.01	0.00	0.01	0.00	0.88	0.02	0.02	0.06
anger	0.09	0.02	0.00	0.00	0.04	0.80	0.04	0.02
sad	0.04	0.06	0.00	0.00	0.03	0.05	0.80	0.01
surprise	0.01	0.06	0.01	0.00	0.02	0.01	0.01	0.88

CONCLUSION

COMPARISION

		F1-Score			
classes	images	model1	model2	model3	model4
disgust	356	0.708	0.666	0.744	0.747
fear	393	0.669	0.74	0.718	0.761
neutral	360	0.914	0.906	0.772	0.919
happy	764	0.95	0.955	0.929	0.977
contempt	368	0.826	0.815	0.87	0.88
anger	400	0.713	0.808	0.748	0.795
sad	346	0.78	0.697	0.725	0.803
surprise	554	0.859	0.844	0.785	0.884
macro avg (Recall)	3541	0.802	0.804	0.786	0.846
macro avg (F1-score)	3541	0.803	0.806	0.786	0.847
Top-2-Accuracy	3541	0.932	0.934	0.922	0.948
accuracy	3541	0.82	0.823	0.802	0.862

- The ensemble performed significantly better than the individual models.
- EfficientNet B0 lived up to its name, being more efficient in terms of both parameters and computational resources, while making fewer compromises on performance.
- DenseNet121 outperformed the other individual models but was quite resource-intensive in terms of computational power.
- VGGFace also delivered strong performance but was demanding both in terms of computational power and the number of parameters it used.

REFRENCES

Dataset :

AFFECTNET - DATASET LINK :
HTTPS://WWW.KAGGLE.COM/DATASETS/NOAMSEGAL/AFFECTNET-TRAINING-DATA?SELECT=HAPPYGRAPH TEXT

Papers :

Ichimura, T. and Kamada, S., 2020, September. A distillation learning model of adaptive structural deep belief network for affectnet: facial expression image database. In 2020 9th International Congress on Advanced Applied Informatics (IIAI-AAI) (pp. 450-455). IEEE.

THANK YOU